Prediction of Real-Gas PVT Behaviour, Thermodynamic Departure Functions, and Pure-Component Fugacity Using the Peng–Robinson Equation of State

Author: Harshith N | **Language:** C++17 | **Build:** CMake 3.20+ (MinGW GCC 13.2) | **Date:** November 2026

1. Introduction, Objectives, and Physical Scope

Accurate prediction of phase equilibria, volumetric behaviour, and thermodynamic properties of fluids is fundamental to chemical process engineering. Ideal-gas formulations fail significantly at elevated pressures and near saturation boundaries. The objective of this project is to develop a modular, scientifically rigorous C++17 application implementing the canonical Peng–Robinson (PR-1976) cubic equation of state for pure fluids. The software predicts: (1) Forward and inverse pressure–volume–temperature (PVT) behaviour; (2) Analytical compressibility factor roots with complete mechanical stability classification; (3) Residual enthalpy (H^{res}) and residual entropy (S^{res}); (4) Pure-component fugacity coefficients ($\ln \phi$, ϕ) and fugacities (f); (5) Reference-state total enthalpy and entropy referenced to the ideal gas at 298.15 K and 1.0 bar; and (6) Thermodynamic phase preference and saturation coexistence checks via comparison of phase chemical potentials.

Physical Scope & Data Conventions: The software treats pure chemical components only; multicomponent mixtures and mixing rules are out of scope. Curated property datasets are provided for eight industrial fluids: Methane (CH_4), Ethane (C_2H_6), Propane (C_3H_8), n-Butane (C_4H_{10}), Nitrogen (N_2), Oxygen (O_2), Carbon Dioxide (CO_2), and Water Vapor (H_2O). All calculations are performed internally in strict SI units (Pa, K, m^3/mol, J/mol, J/(mol·K)).

2. Governing Mathematical Model and Thermodynamic Equations

2.1 The Peng–Robinson Cubic Equation of State (PR-1976)

The canonical pressure-explicit cubic formulation proposed by Peng and Robinson (1976) models pressure P as the sum of a repulsive hard-sphere contribution and an attractive intermolecular dispersion contribution:

$$P(T, v) = [RT / (v - b)] - [a(T) / (v^2 + 2bv - b^2)]$$

where R = 8.314462618 J/(mol·K) is the universal gas constant (CODATA 2018), v is molar volume [m^3/mol], b is the co-volume parameter representing finite molecular volume, and $a(T)$ is the temperature-dependent attractive parameter. Enforcing the critical-point inflection constraints $(\partial P/\partial v)_T = 0$ and $(\partial^2 P/\partial v^2)_T = 0$ yields the universal critical parameter relationships:

$$a_c = \Omega_a (R^2 T_c^2 / P_c), \quad b_c = \Omega_b (R T_c / P_c), \quad \text{where } \Omega_a \approx 0.4572355289,\ \Omega_b \approx 0.0777960739$$

Note on PR Parameter Precision: In literature and textbook presentations, Ω_a and Ω_b are commonly rounded to 0.45724 and 0.07780, respectively. The software implementation deliberately employs the full analytical double-precision roots derived directly from the critical-point cubic constraint ($\Omega_b = (\sqrt{2} - 1) / [3(1 + \sqrt{2})]$), ensuring exact mathematical consistency with zero residual at the critical point.

2.2 Temperature Dependence and Analytical Derivative

The attractive parameter accounts for thermal kinetic effects through the dimensionless scaling function $\alpha(T_r, \omega)$:

$$a(T) = a_c \alpha(T), \quad \alpha(T) = [\,1 + \kappa (1 - (T / T_c)^{1/2})\,]^2, \quad \kappa = 0.37464 + 1.54226\,\omega - 0.26992\,\omega^2$$

where $T_r = T / T_c$ is the reduced temperature and ω is the Pitzer acentric factor. The analytical temperature derivative (da/dT), essential for evaluating departure functions, is evaluated exactly as:

$$da/dT = -[\,a_c \kappa \alpha^{1/2}(T)\,] / [\,(T T_c)^{1/2}\,] \quad [Pa·m^6/(mol^2·K)]$$

2.3 Dimensionless Compressibility Polynomial

Defining the compressibility factor $Z \equiv Pv / (RT)$ and the dimensionless parameters $A \equiv aP / (RT)^2$ and $B \equiv bP / (RT)$, the Peng–Robinson EOS transforms into a monic cubic polynomial:

$$Z^3 - (1 - B) Z^2 + (A - 3B^2 - 2B) Z - (AB - B^2 - B^3) = 0$$

At physically meaningful fluid states, only roots satisfying $Z > B$ correspond to molar volumes exceeding the co-volume ($v > b$). Any mathematical roots with $Z \le B$ represent unphysical negative or sub-molecular volumes and are strictly filtered out.

2.4 Residual Thermodynamic Departure Functions

Residual departure functions describe the difference between the actual thermodynamic property of a fluid and that of an ideal gas at the same temperature and pressure: $M^{res}(T, P) \equiv M(T, P) - M^{ig}(T, P)$. Integrating along an isothermal path from infinite volume (ideal gas limit) to molar volume v using the exact Maxwell relations yields the closed-form PR expressions:

$$H^{res}(T, P, Z) = RT\,(Z - 1) + [\,(T\,da/dT - a) / (2\sqrt{2}\,b)\,] \cdot \ln [\,(Z + (1 + \sqrt{2})B) / (Z + (1 - \sqrt{2})B)\,] \quad [J/mol]$$

$$S^{res}(T, P, Z) = R \ln(Z - B) + [\,(da/dT) / (2\sqrt{2}\,b)\,] \cdot \ln [\,(Z + (1 + \sqrt{2})B) / (Z + (1 - \sqrt{2})B)\,] \quad [J/(mol \cdot K)]$$

Within the mathematical framework of the Peng–Robinson EOS, these analytical expressions satisfy the fundamental thermodynamic identity $G^{res} = H^{res} - T\,S^{res} = RT \ln \phi$ identically.

2.5 Pure-Component Fugacity Coefficient and Fugacity

The fugacity coefficient $\phi \equiv f / P$ represents the dimensionless activity of the fluid relative to an ideal gas. From the classical thermodynamic definition $\ln \phi = \int_0^P [(Z - 1) / P]\,dP$, the exact PR analytical solution is:

$$\ln \phi = (Z - 1) - \ln(Z - B) - [\,A / (2\sqrt{2}\,B)\,] \cdot \ln [\,(Z + (1 + \sqrt{2})B) / (Z + (1 - \sqrt{2})B)\,], \quad f = \phi \cdot P \quad [Pa]$$

In the ideal gas limit ($P \to 0$), $Z \to 1$, $B \to 0$, $\ln \phi \to 0$, and $\phi \to 1.0$ identically.

2.6 NIST Shomate Heat Capacity and Reference-State Total H and S

To calculate absolute state enthalpy $H(T, P)$ and entropy $S(T, P)$, the program integrates the authentic 5-parameter NIST Shomate isobaric heat capacity correlation for the ideal gas state ($t = T / 1000$):

$$C_p^{\circ}(t) = A + Bt + Ct^2 + Dt^3 + E / t^2 \quad [J/(mol \cdot K)]$$

Exact analytical integration yields the temperature departures ΔH^{ig} and ΔS^{ig}:

$$\Delta H^{ig}(T_{ref} \to T) = 1000\,[\,At + Bt^2/2 + Ct^3/3 + Dt^4/4 - E/t\,]_{t_{ref}}^{t} \quad [J/mol]$$

$$\Delta S^{ig}(T_{ref} \to T) = [\,A \ln t + Bt + Ct^2/2 + Dt^3/3 - E / (2t^2)\,]_{t_{ref}} \quad [J/(mol \cdot K)]$$

Reference State Convention: Total enthalpy and entropy are defined relative to the ideal gas reference state at T_{ref} = 298.15 K (25.0 °C) and P_{ref} = 100,000 Pa (1.0 bar), where $H_{total}(T_{ref}, P_{ref}) \equiv 0.0$ J/mol and $S_{total}(T_{ref}, P_{ref}) \equiv 0.0$ J/(mol·K). The total properties at arbitrary (T, P, Z) are:

$$H(T, P, Z) = \Delta H^{ig}(T_{ref} \to T) + H^{res}(T, P, Z) - H^{res}(T_{ref}, P_{ref}, Z_{ref})$$

$$S(T, P, Z) = \Delta S^{ig}(T_{ref} \to T) - R \ln(P / P_{ref}) + S^{res}(T, P, Z) - S^{res}(T_{ref}, P_{ref}, Z_{ref})$$

Steam Reference State Limitation: NIST Shomate gas-phase parameters for Water Vapor are valid exclusively in [500.0, 1700.0] K. Because T_{ref} = 298.15 K lies outside this interval, total H and S relative to standard conditions are reported as N/A with an explicit diagnostic message rather than extrapolating invalid polynomials. Within the PR model, PVT, fugacity, and residual departure properties for water remain fully operational across all valid temperatures.

2.7 Low-Pressure Taylor Series Regularization

At near-vacuum pressures ($P < 10$ Pa), $B \to 0$, causing the logarithmic quotient term $I_L = \ln[(Z + (1+\sqrt{2})B)/(Z + (1-\sqrt{2})B)] / (2\sqrt{2}\,B)$ to approach 0 / 0. In the C++ implementation, this is regularized by defining $u \equiv 2\sqrt{2}\,B / [Z + (1 - \sqrt{2})B]$ and expanding $\ln(1 + u)/u$:

$$I_L = [\,1 / (Z + (1 - \sqrt{2})B)\,] \cdot [\,1 - u/2 + u^2/3 - u^3/4\,] \quad for\ |u| < 10^{-5}$$

This maintains full double-precision numerical continuity without catastrophic subtraction cancellation down to $P = 10^{-5}$ Pa.

2.8 Analytical Cubic Root Solver and Stability Classification

The cubic polynomial in Z is solved analytically using Cardano's method transformed into depressed form $y^3 + py + q = 0$ via the Tschirnhaus substitution $Z = y - c_2/3$. The cubic discriminant $\Delta = q^2/4 + p^3/27$ determines the physical root topology: (1) If $\Delta > 0$, exactly one real root exists (Cardano's real radical formula); (2) If $\Delta < 0$, three distinct real roots exist (Vieta's trigonometric formulation); and (3) If $\Delta = 0$, multiple real roots coalesce.

Mechanical Stability Criterion: For every physical root ($Z > B$), the isothermal pressure–volume derivative $(\partial P/\partial v)_T$ is evaluated analytically from the PR equation: $(\partial P/\partial v)_T = -RT/(v - b)^2 + a(T)(2v + 2b)/(v^2 + 2bv - b^2)^2$. Roots with $(\partial P/\partial v)_T < 0$ are **STABLE** (compression increases pressure). Roots with $(\partial P/\partial v)_T \geq 0$ are **UNSTABLE** (spinodal states; physically unobservable).

Phase Preference Criterion & Equilibrium: In the three-root subcritical region, the smallest root (Z_{min}) is liquid-like, the largest root (Z_{max}) is vapour-like, and the intermediate root is mechanically unstable. Thermodynamic phase stability is governed by the minimum Gibbs free energy, which corresponds to the lower fugacity coefficient ϕ: $\Delta G^{res} = G_{liq}^{res} - G_{vap}^{res} = RT \ln(\phi_{liq} / \phi_{vap})$. If $\phi_{vap} < \phi_{liq}$ ($\Delta G^{res} > 0$), the vapour phase has lower chemical potential and is the preferred stable state. If $\phi_{liq} < \phi_{vap}$ ($\Delta G^{res} < 0$), the liquid phase is preferred. At user-specified states where multiple roots exist, if the relative difference $|\phi_{liq} - \phi_{vap}| / \phi_{vap} \leq 10^{-5}$, the program identifies approximate saturated two-phase equilibrium. Note that the program performs state evaluation at specified (T, P) inputs rather than iteratively solving for saturation pressure via a bubble/dew-point algorithm from scratch.

3. Software Architecture and Implementation

The software is designed as a modular, header/source C++17 library (thermo_core) strictly decoupled from user-interface logic. It relies exclusively on the C++ standard library (<cmath>, <vector>, <string>, <optional>, <iostream>) with zero external dependencies.

Module / Component	Source Files	Architectural Responsibility & Function
Constants	include/thermo/Constants.hpp	Universal gas constant R (CODATA 2018), reference-state constants, exact analytical PR-1976 parameters.
Component & Database	src/Component.cpp src/ComponentDatabase.cpp	Validates physical parameters; stores curated database for 8 industrial gases; evaluates analytical Shomate integrals.
Cubic Solver	src/CubicSolver.cpp	Analytical Cardano/Vieta polynomial solver; filters duplicate roots; guarantees residuals $\leq 10^{-12}$.
PR EOS Engine	src/PengRobinsonEOS.cpp	Forward/inverse PVT, $(\partial P/\partial v)_T$, departure functions H^{res}, S^{res}, fugacity ϕ, and phase preference.
Interactive CLI	src/cli/CLIController.cpp src/cli_main.cpp	Stream-decoupled interactive menu (`thermo_cli`), unit conversions (K/°C, Pa/bar/MPa), benchmark demo runner.
Automated Test Suite	tests/test_*.cpp (5 targets)	373 automated assertions covering cubic roots, Shomate quadrature, PVT reciprocity, Maxwell derivatives, and CLI.

3.1 Build System and Execution Instructions

The project is built using CMake 3.20+ with strict compiler warnings (`-Wall -Wextra -Wpedantic`):

cmake -B build -S . -DCMAKE_BUILD_TYPE=Release && cmake --build build

To execute the full automated CTest suite (5 executables, 373 assertions):

ctest --test-dir build --output-on-failure

To run the interactive CLI or automated demonstration benchmarks:

./build/thermo_cli # Interactive mode
./build/thermo_cli --demo all # Automated demonstration benchmarks

4. Verification Results and Audited Numerical Benchmarks

The implementation was evaluated using direct benchmark demonstrations and independent mathematical consistency checks. The table below reports values confirmed by direct execution of `thermo_cli`:

Case / State	Fluid & Phase	Z (PR-1976)	v [m^3/mol]	ϕ (Fugacity Coeff)	H^{res} [J/mol]	S^{res} [J/(mol·K)]	Status / Preference
Demo 1 300 K, 5.0 MPa	Methane (Gas)	0.901844	4.4990×10^{-4}	0.901385 (f=4.507 MPa)	–902.34	–2.1446	Single Phase (Stable)
Demo 2 300 K, 7.0 MPa	CO_2 (Dense Fluid)	0.198057	7.0575×10^{-5}	0.638285 (f=4.468 MPa)	–9374.37	–27.5150	Single Phase (Stable Liquid-like)
Demo 3 (Root 1) 320 K, 1.0 MPa	Propane (Liquid-like)	0.036453	9.6988×10^{-5}	1.239280 (f=1.239 MPa)	–14,854.34	–48.2035	Liquid Branch (Metastable)
Demo 3 (Root 2) 320 K, 1.0 MPa	Propane (Intermediate)	0.091826	2.4432×10^{-4}	1.385615 (f=1.386 MPa)	–8,210.60	–28.3698	**UNSTABLE** ($\partial P/\partial v > 0$, Spinodal)
Demo 3 (Root 3) 320 K, 1.0 MPa	Propane (Vapour-like)	0.850555	2.2630×10^{-3}	0.868006 (f=0.868 MPa)	–1,131.04	–2.3575	**VAPOUR PREFERRED** (ΔG^{res} = +947.42 J/mol)

4.1 In-Depth Analysis of Multi-Root Propane Behaviour (Demo 3)

At 320 K and 1.0 MPa, Propane lies in the subcritical three-root region of the Peng–Robinson equation ($\Delta = -1.083\times10^{-5} < 0$), generating three real roots: Root 1 (liquid-like, Z = 0.036453), Root 2 (intermediate, Z = 0.091826), and Root 3 (vapour-like, Z = 0.850555). The existence of three physical roots does not alone prove vapour–liquid coexistence; rather, at fixed temperature and pressure, phase preference is determined by comparing the fugacity coefficients of the mechanically stable roots. The intermediate root exhibits $(\partial P/\partial v)_T = +1.8825\times10^{10}$ Pa/(m^3/mol) > 0, confirming it is mechanically unstable and located inside the spinodal zone. For the two mechanically stable roots, $\phi_{vap} = 0.868006 < \phi_{liq} = 1.239280$, producing $\Delta G^{res} = RT\ln(\phi_{liq}/\phi_{vap}) = +947.42$ J/mol > 0. The vapour phase possesses strictly lower chemical potential ($\mu_{vap} < \mu_{liq}$) and represents the thermodynamically preferred stable state.

4.2 Automated Testing and Independent Differential Maxwell Cross-Checks

The repository contains an automated test suite of **373 assertions across 5 CTest executables (100% pass rate)**. Circularity in validation was avoided by implementing independent differential checks: (1) **Gibbs–Helmholtz Cross-Check:** H^{res} was verified against numerical central differences $-RT^2\,(\partial \ln\phi/\partial T)_P$ with δT = 0.1 mK, matching to a relative error of 3.8×10^{-10}; (2) **Pressure Fugacity Differential:** $(\partial \ln\phi/\partial P)_T$ was matched to $(Z-1)/P$ to within 1.4×10^{-9}; and (3) **Numerical Quadrature:** Analytical Shomate integrals for ΔH^{ig} and ΔS^{ig} were cross-checked against an independent 1000-subinterval composite Simpson's 1/3 quadrature to within 10^{-6}.

5. Assumptions, Limitations, and Conclusion

Model Scope and Physical Limitations: (1) *Pure Fluids Only:* Multicomponent mixtures and mixing rules are out of scope. (2) *Steam Reference State Limitation:* NIST Shomate gas-phase parameters for Water Vapor are valid only in [500.0, 1700.0] K. Because T_{ref} = 298.15 K lies outside this interval, total H and S are reported as N/A with clear diagnostics; PVT, fugacity, and residual departure functions remain fully operational within the PR model framework. (3) *No Unphysical Extrapolation:* Calculating outside documented Shomate intervals throws exceptions rather than generating erroneous values. (4) *Cubic Liquid Volume Accuracy:* Near the critical point, two-parameter cubic EOSs typically underestimate liquid densities by 5–10% without volume translation (Peneloux).

Conclusion: The software provides a mathematically sound, verified, and robust engineering implementation of the pure-component Peng–Robinson equation of state. It successfully evaluates volumetric behaviour, multi-root phase preference, and thermodynamic departure functions while maintaining high numerical precision.

References

[1] Peng, D.-Y., & Robinson, D. B. (1976). A New Two-Constant Equation of State. *Ind. Eng. Chem. Fundam.*, 15(1), 59–64. https://doi.org/10.1021/i160057a011

[2] Linstrom, P. J., & Mallard, W. G., Eds. (2023). *NIST Chemistry WebBook, NIST Standard Reference Database Number 69*, National Institute of Standards and Technology, Gaithersburg MD. https://webbook.nist.gov

[3] Poling, B. E., Prausnitz, J. M., & O'Connell, J. P. (2001). *The Properties of Gases and Liquids*, 5th Ed., McGraw-Hill, New York.